



Insect Biomass & Red-Listed Species

In Bavaria, Germany

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Introduction



Malaise traps were set out on land plots to collect insects, the biomass of these collected insects and the number of red-listed (threatened) species was recorded.

Landscape Types

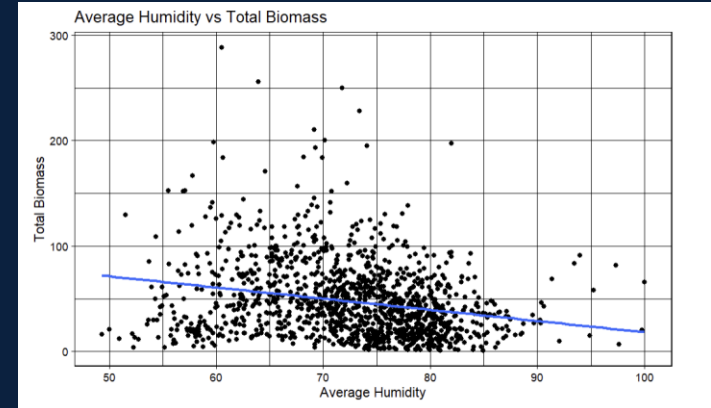
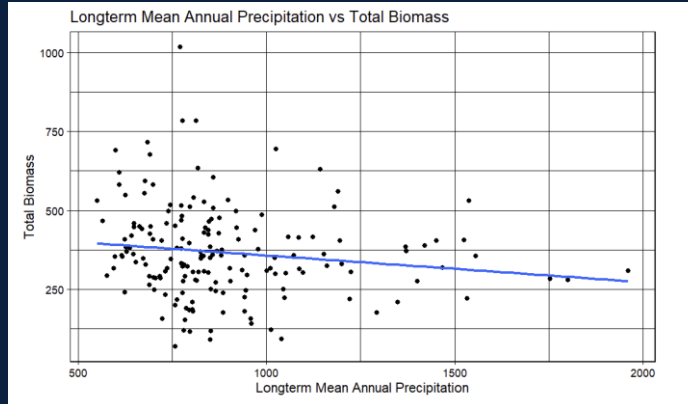
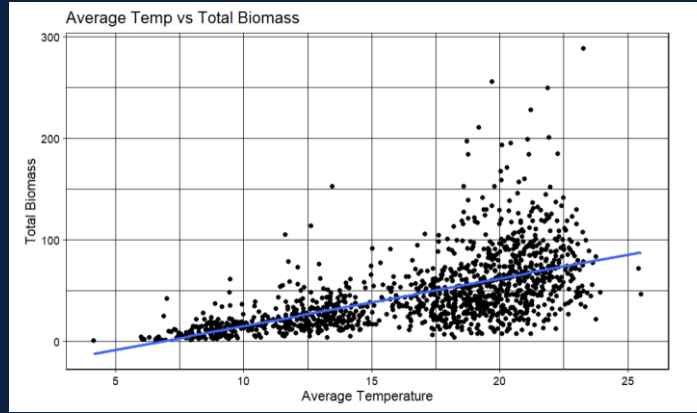
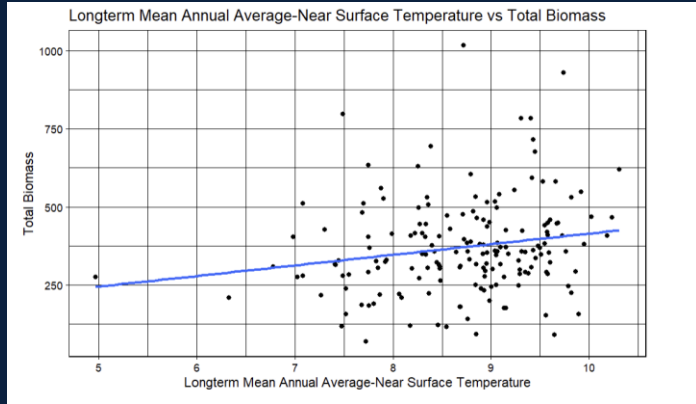
- Agricultural
- Urban
- Semi-Natural



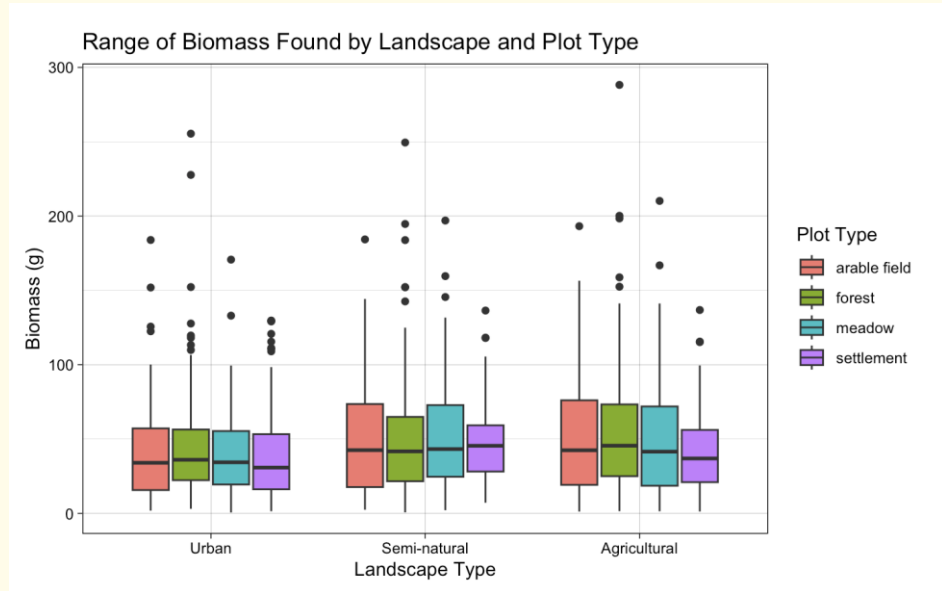
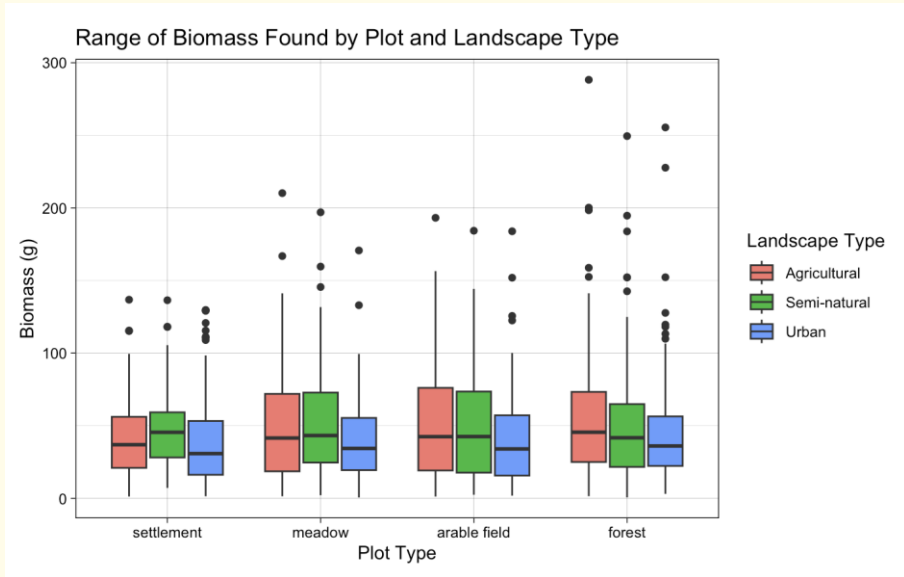
Plot Types

- Forest
- Meadow
- Arable Field
- Settlement

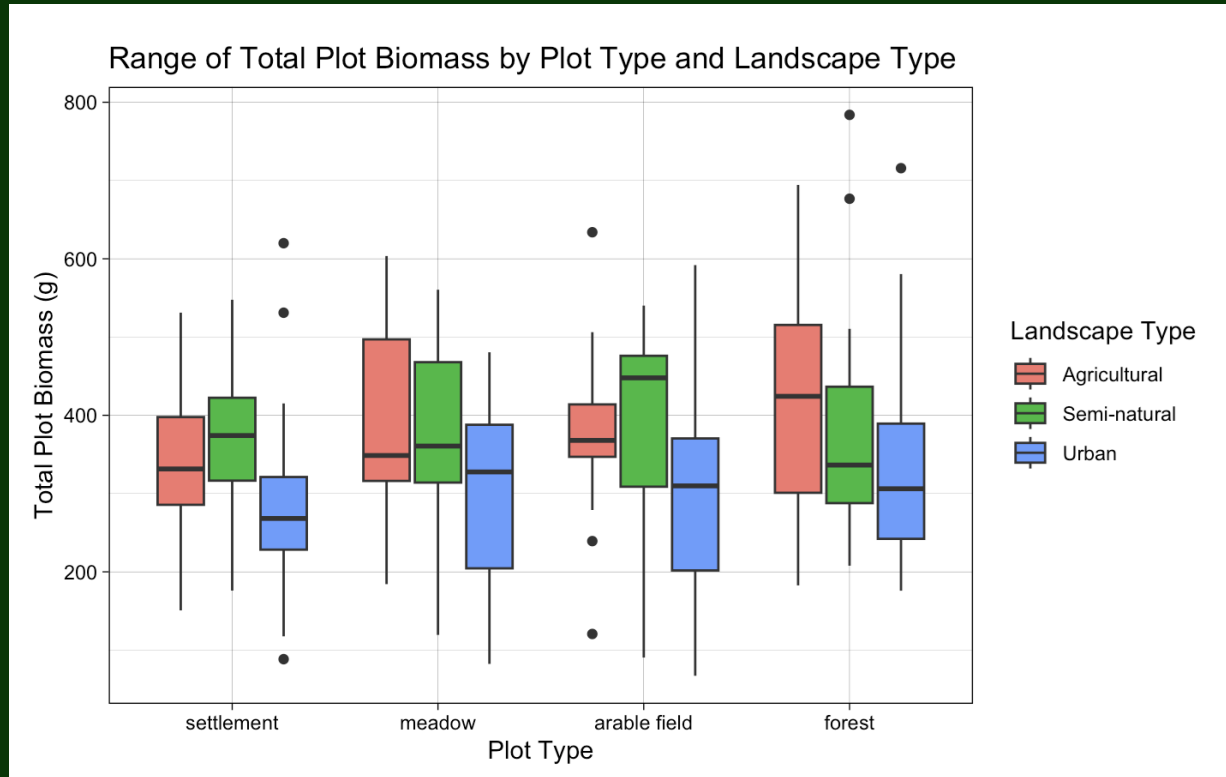
Comparing Biomass to Temp/Humidity



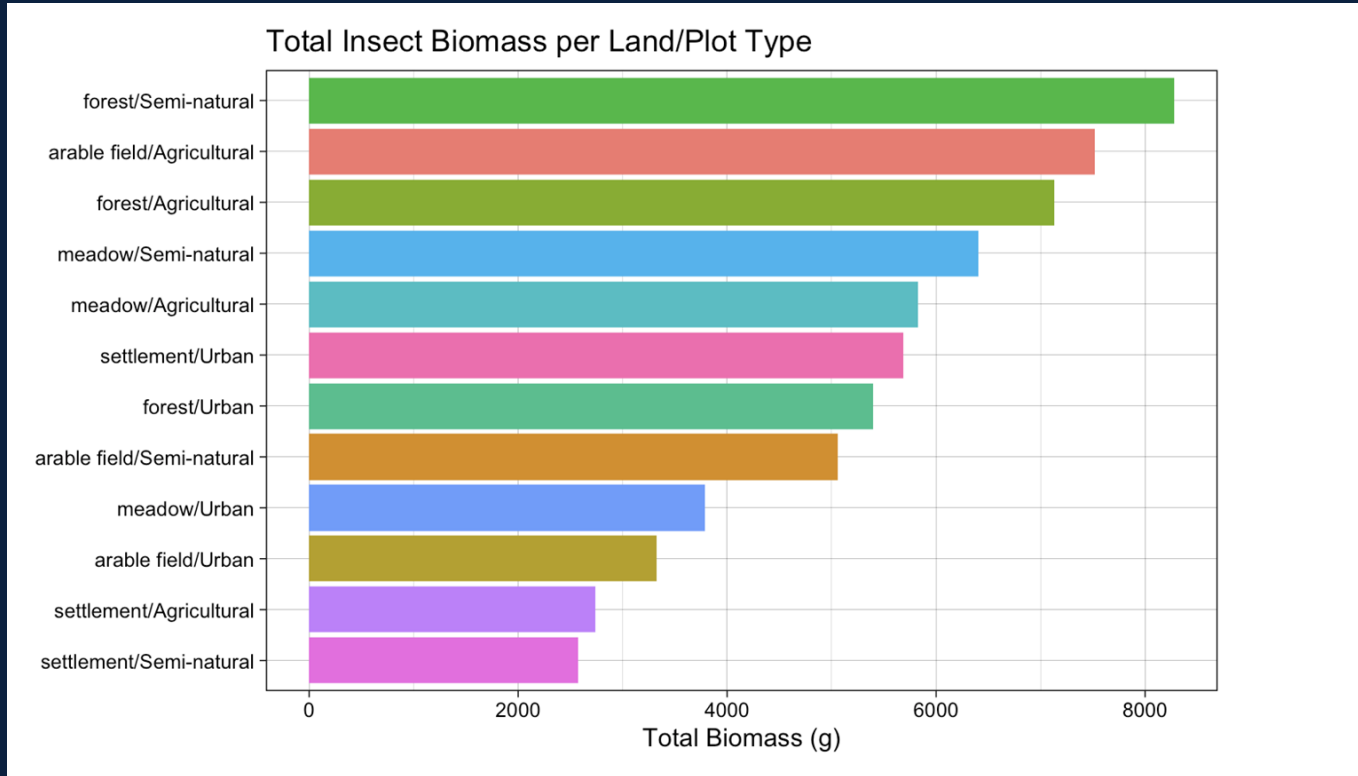
Range of Biomass Collected Across Plot and Landscape Types



Range of Total Plot Biomass Across Plot and Landscape Types

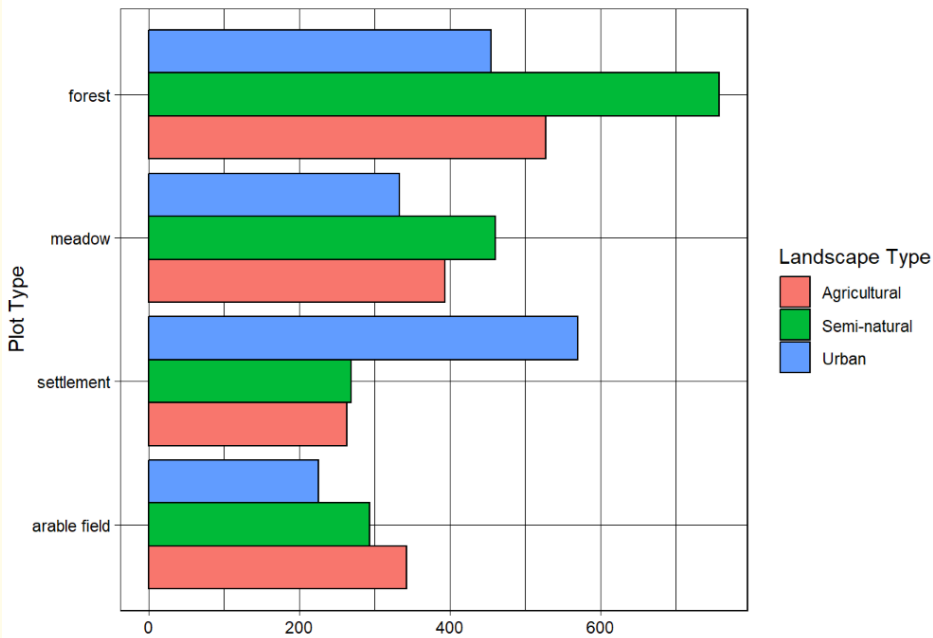


Total Insect Biomass Per Land/Plot Type

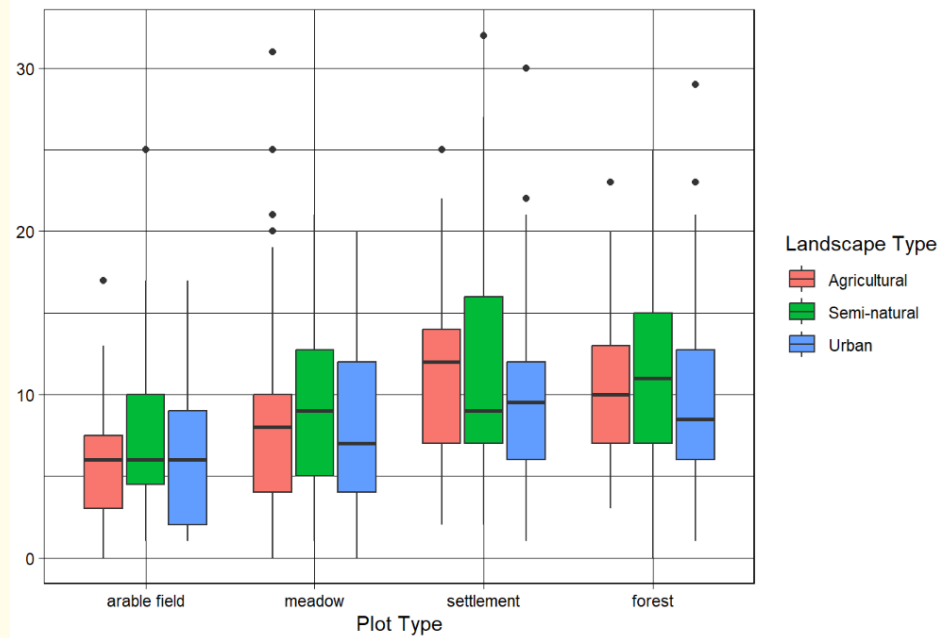


No. of Red-Listed Species by Plot and Landscape Type

No. of Red-Listed Species Found by Plot Type & Landscape



No. of Red-Listed Species Found by Plot Type & Landscape



Conclusion

- No obvious hidden causal factors from weather.
- Urban areas had the lowest total biomass but agricultural land & especially arable fields had the lowest no. of red-listed species.
- Semi-natural forests had the highest total biomass & no. of red-listed species.

Shiny App

Take A Closer Look!

Select Landscape Type:

- ☒ All ☐ Agricultural
☐ Semi-natural ☐ Urban

Select Plot Type:

- ☒ All ☐ settlement
☐ meadow ☐ arable field
☐ forest

Select Plot:

5636_4_A



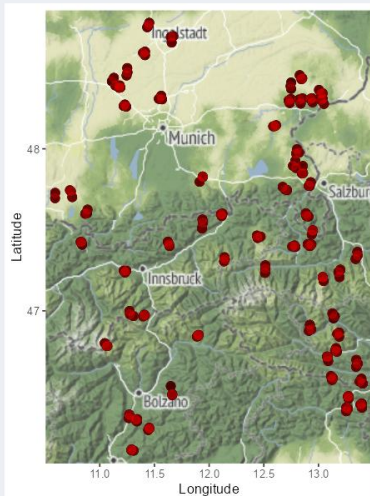
Total Grams of Biomass:

367.86

Total No. of Red-Listed Species:

26

Map:



Source: Uhler, J., et. al.; Relationship of insect biomass and richness with land use along a climate gradient. Nature Communications 12 October 2021; 12: 5946.
<https://doi.org/10.1038/s41467-021-26181-3>